

[WORKING TITLE — see PROMPT 19 E2]

On the Evaluation of Structural Priors in Unrolled Channel Estimation for Rydberg Atomic Receivers

Abdullah Haatim

Abstract—[ABSTRACT IS A STUB — PROMPT 19 E2.] It cannot be written until batch C2 settles which of the three P29 outcomes fires, because the three are three different papers and the abstract is the sentence that commits to one. What it must contain either way: the gradient-concentration finding (89.7% of the gradient below 5 dB, a per-bin share spanning a factor of 30.0); the matched-adequacy protocol; the corrected loss and its +2.171 dB three-seed gain at $\text{SNR} \geq 5$ dB with no low-SNR cost; the pilot-efficiency distinction at 2.231 dB; and simulation-only scope. What it must not contain until C2 reports: any scalar for the structural gain.

Index Terms—Algorithm unrolling, channel estimation, evaluation methodology, phase retrieval, Rydberg atomic receiver, structural priors.

I. INTRODUCTION

UNROLLING an iterative estimator into a trained network [1], [2] has become standard for channel estimation, including for Rydberg atomic receivers, where the magnitude-only readout makes the problem one of phase retrieval [3], [4]. A recurring next step is to add an explicit structural prior — a low-rank projection, a sparsity constraint — inside the unrolled loop, and to report the resulting improvement as evidence that the prior supplies information the network lacks.

This paper argues that such evidence is usually not decisive, and that the reason is a convention so widespread it is rarely stated: the training loss is a per-sample normalized error, averaged over a training set spanning a wide SNR range.

This is a finding about evaluation practice, and we did not arrive at it as critics. We built a Hankel-structured unrolled estimator, measured an advantage at high SNR, and believed it. The control described in Sec. III was run to characterize that advantage, not to remove it. The first casualty of the protocol we propose was our own method, and we report it because a result that survives an author's attempt to keep it is worth more than one that was never tested.

Contributions.

- 1) An attribution decomposition of an unrolled estimator's gain, separating the learned filter, the attention block, and unrolling itself against a matched non-unrolled control.
- 2) Identification of the confound: per-sample normalized error over mixed SNR concentrates the gradient at

low SNR, so any mechanism that improves high-SNR behaviour will appear to add information. We give a matched-adequacy protocol that separates the two.

- 3) A corrected loss that is not a workaround but a fix, gaining +2.171 dB at $\text{SNR} \geq 5$ dB over three seeds with no low-SNR penalty — indeed with a gain in every bin of every seed, including the two bins the weighting penalizes most.
- 4) A demonstration that pilot efficiency and pilot-count generalization are distinct and are conflated by a single reported curve.
- 5) [STUB] The structural-prior contribution, pending C2.

[STUB — INTRODUCTION, structural-prior claim]

One paragraph stating what the control did to Δ_H , and the corresponding contribution item. It cannot be written until batch C2 reports, because the three P29 outcomes give three different claims: a stable gain whose value depends on training adequacy; a quantity with no stable sign under any design; or a stable gain materially smaller than first reported. What is already established and belongs here whichever fires: under focused training Δ_H has no stable sign across three seeds, and the per-bin ordering — a cost at low SNR, a benefit at high SNR — replicates on every arm measured while the SNR at which it crosses does not.

II. SYSTEM MODEL AND ESTIMATOR

An N -element uniform linear array serves K users over P pilots. With $\mathbf{G} \in \mathbb{C}^{N \times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the receiver observes magnitudes only,

$$\mathbf{Z} = |\mathbf{G}\mathbf{S} + \mathbf{B} + \mathbf{W}|, \quad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2). \quad (1)$$

Each user's channel is a sum of L_k plane waves on the array manifold, with $L_k \sim \mathcal{U}\{3, 7\}$ [5] and angles drawn uniformly on $(-90^\circ, 90^\circ)$ (Table I); this is a geometric ULA model, not a clustered one. The classical baseline is an expectation-maximization Gerchberg-Saxton iteration (EM-GS) [6]; the unrolled estimator ("URformer") replaces T iterations with T learned layers, each a data-consistency step followed by a learned filter, a gate, and a Transformer block across users [3].

We implement this estimator faithfully on the geometric ULA model of (1) rather than reproducing any published result; the operating point differs. Nothing here should be read as a reproduction or a validation of prior work, nor as a critique of any specific paper. The convention we examine is near-universal, and our own estimator uses it.

A. Haatim is with the Department of Electrical and Electronics Engineering, Birla Institute of Technology and Science, Pilani — Pilani Campus, India (e-mail: f20220519@pilani.bits-pilani.ac.in).

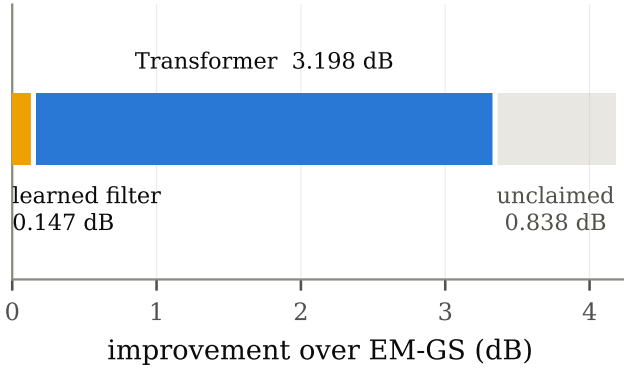


Fig. 1. Attribution of the unrolled estimator’s gain over EM-GS at 80k training samples. “Unclaimed” is the remainder of the genie-aided least-squares headroom, which is a reference and not a bound.

The reference curve is genie-aided. Where a reference appears, it is an unstructured least-squares estimate formed from the *true* phase of the noiseless field. It is not a bound: it presumes information no receiver has, and being restricted to unstructured least squares it can be passed by an estimator that exploits channel structure — which we in fact observe below 5 dB in Sec. IV-A.

III. ATTRIBUTION AND THE MATCHED-ADEQUACY PROTOCOL

A. Where the gain comes from

Fig. 1 decomposes the unrolled estimator’s 3.345 dB improvement over EM-GS. The learned per-layer filter, 980 parameters, contributes 0.147 dB; the attention block contributes the remaining 3.198 dB, or 96%. The filter’s contribution does not grow with data — it falls slightly, $0.193 \rightarrow 0.147$ dB from 20k to 80k training samples — which is what a never-capacity-limited 980-parameter component should do.

Unrolling is doing real work independently of attention. Against a matched control that applies one Transformer block to a converged EM-GS estimate rather than unrolling, the unrolled estimator gains 0.920 dB, CI $[+0.888, +0.988]$, winning on 86.5% of trials.

B. The confound

The training loss is the per-sample normalized error $\|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 / \|\mathbf{G}\|_F^2$, averaged over a batch drawn across the whole SNR range. Because the normalized error of a low-SNR realization is orders of magnitude larger than that of a high-SNR one, this average is dominated by the low-SNR tail. Measured directly on our estimator, 89.7% of the gradient comes from trials below 5 dB, and the per-bin gradient share spans a factor of 30.0 across the range (Fig. 2, upper panel).

The consequence is that the high-SNR regime is systematically underfitted, and *any* mechanism that improves high-SNR behaviour will appear to add information. A structural prior is exactly such a mechanism.

TABLE I
CONFIGURATION.

Parameter	Value
Array elements N / users K / pilots P	32 / 3 / 20
Paths per user L_k	$\mathcal{U}\{3, 7\}$ [5]
SNR (train and test)	$\mathcal{U}[-10, 20]$ dB
Reference-to-signal ratio	10 dB
Unrolled layers T	10
Trainable parameters	1,586,900
Training samples / epochs	80,000 / 13
Test realizations	2,000 (paired)

C. The protocol

We propose the following control, which is cheap and which we recommend be run before any structural prior is credited.

- 1) Measure the per-bin gradient share of the training loss. If it is far from uniform, the regime receiving least gradient is underfitted.
- 2) Retrain *both* the structured and unstructured arms with training concentrated on the regime of interest, matching data budget, schedule, seed and initialization.
- 3) Re-measure the contrast. What survives is attributable to the prior; what does not was compensating for training inadequacy.

[STUB — RESULT OF APPLYING THE PROTOCOL]

The paragraph reporting what the control did to Δ_H , and its figure. Waiting on batch C2, which measures the seed spread of the mixed-SNR contrast; the focused contrast is already measured on three seeds. No scalar may be quoted here until then, and the figure must be rebuilt from C2 and C3 rather than repurposed — the retired one plotted a single seed of each design as though the pair were a measurement.

Scope of this negative result. It holds at $N = 32$, $K = 3$, this data budget, and the two pilot counts tested. It is not a general claim that structural priors cannot help unrolled estimators. Indeed the same prior applied to the *classical* estimator, where no training adequacy is at stake, gives a large and stable gain, and under a distribution shift in path richness the structured network degrades least (+0.48 dB against +0.73 dB). The prior buys robustness here; it is the in-distribution accuracy claim that does not survive the control.

IV. NUMERICAL RESULTS

Table I lists the configuration. The reported statistic is the paired per-trial median within an SNR bin, with bootstrap 95% confidence intervals over 2000 paired test realizations.

A. A corrected loss, not a workaround

The confound is a property of the loss, so we fix the loss. We weight each sample by a static per-bin factor $w(b) = c/m(b)$, where $m(b)$ is the mean per-sample normalized error in bin b measured once before training, with c set so the weights have unit mean. This requires no schedule, no tuning, and no change to the architecture.

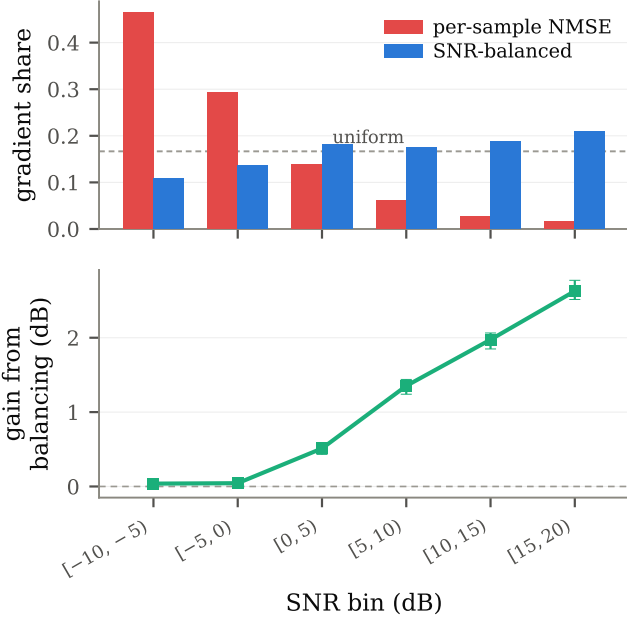


Fig. 2. Cause and effect. Upper: per-bin gradient share under the per-sample normalized loss and under SNR balancing. Lower: the resulting per-bin gain, paired per-trial median with bootstrap 95% CI over test realizations, on the seed for which the prediction was registered.

It works as designed. The realized per-bin gradient share spans a factor of 1.9 after balancing against 30.0 before, and the share below 5 dB falls from 0.897 to 0.427 against an ideal of 0.5 — a slight over-correction.

The estimation result is better than we predicted, and it is measured on three seeds rather than one. Against identically-trained unbalanced baselines, balancing gains +1.902, +2.242 and +2.369 dB at $\text{SNR} \geq 5$ dB, a mean of +2.171 dB. Each seed’s bootstrap 95% confidence interval *over test realizations* excludes zero. The spread *over seeds* is a separate quantity and is reported as one: its standard deviation is 0.241 dB, on three seeds and therefore two degrees of freedom, so it is indicative rather than precise. Taking the conservative combination, mean $-2\text{SD} = +1.689$ dB, the effect is separated from zero across seeds.

We had pre-registered a low-SNR *cost* of -0.10 to -0.40 dB on the reasoning that the lowest bin’s weight falls about eighteenfold. There was no cost. On the seed for which the prediction was registered the low-SNR change is +0.135 dB, CI over test realizations [+0.101, +0.172]; across the three seeds the two lowest bins gain +0.052 and +0.202 dB on average. Downweighting the bins that dominated the gradient did not harm them, because they were over-served rather than merely well-served.

The gap to the genie-aided reference at $\text{SNR} \geq 5$ dB falls from +2.897 to +1.000 dB. Below 5 dB the balanced estimator passes that reference by up to 1.99 dB, which is consistent rather than anomalous: the reference is restricted to unstructured least squares, and an estimator that has learned the channel geometry is not so restricted.

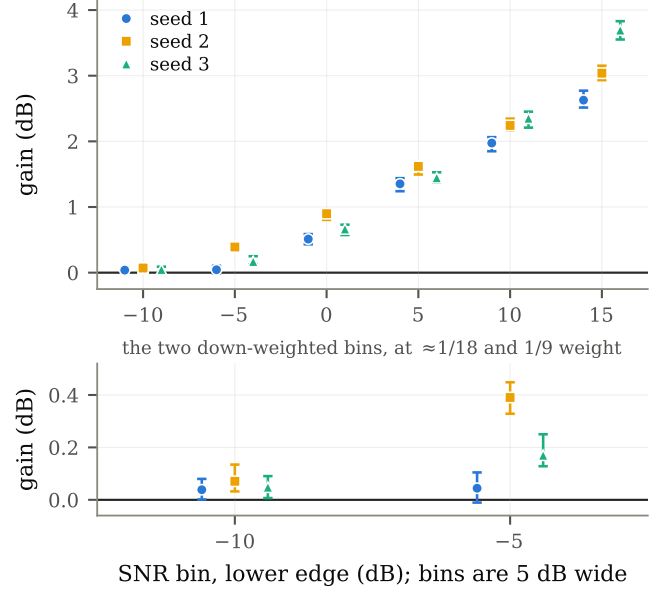


Fig. 3. Every bin, every seed. The gain from SNR balancing, paired per-trial median with bootstrap 95% CI over test realizations, for each of three training seeds. All eighteen cells are positive, including the two lowest bins, whose weights the balancing divides by about 18 and 9.

B. The correction improves the regime it penalizes

The obvious objection to a reweighting is that it trades one regime for another: whatever the high-SNR bins gain, the down-weighted low-SNR bins must have paid. Here they did not, and the check is exhaustive rather than pooled.

Fig. 3 plots every SNR bin for every seed. **All eighteen bin-by-seed cells favour the balanced loss**, and seventeen of the eighteen have a bootstrap 95% confidence interval over test realizations that excludes zero; the single exception is seed 1’s $[-5, 0)$ dB bin at +0.044 dB, CI $[-0.011, +0.105]$. The three-seed means rise monotonically across the range, from +0.052 dB in $[-10, -5)$ to +3.120 dB in $[15, 20)$.

The two lowest bins are the ones that matter for the objection, because the weighting divides their contribution by about 18 and 9 relative to unit mean. They gain anyway. The correction is therefore not a transfer between regimes: it improves the quantity it was weighted away from.

We offer a reading of why, as a hypothesis and not as a result. Under the per-sample loss the gradient was so dominated by the low-SNR tail that the network was underfitted *everywhere*, spending capacity on realizations from which little is recoverable. Rebalancing does not move performance from one regime to another; it stops wasting it. Testing that claim directly would require an ablation we have not run.

C. The structural prior under matched adequacy

[STUB — STRUCTURAL-PRIOR SECTION]

The whole subsection, pending batches C2 and C3. Three things are already established and will survive whichever P29 outcome fires, and they are recorded here so the section is written from them rather than around them. First, under

focused training Δ_H has no stable sign: three seeds give +0.078, +0.057 and -0.076 dB, every bootstrap CI over test realizations excluding zero, and the across-seed SD is 0.083 dB — small, so this is a statement about the effect and not about noise. Second, the paired per-trial median and the ratio of sums disagree in sign, and the reason is measured: the sum of squared errors is dominated by low-SNR realizations, which is exactly where the prior costs accuracy. It is not the abstention mechanism of the classical estimator — the projection is active on every layer of every one of 2000 trials on every seed. Third, the per-bin ordering, a cost at low SNR turning into a benefit at high SNR, replicates on every arm measured, across two initializers, two training designs, two losses and four seeds; the SNR at which it crosses does not, moving across three bin boundaries. The reportable object is that ordering, not any scalar.

D. Pilot efficiency is not pilot-count generalization

A pilot-count curve for a trained estimator can mean two different things: one model trained at some P_0 and evaluated at each P (generalization), or a model trained at each P (efficiency). These are not close. Trained at $P = 10$, the estimator beats the $P = 20$ model evaluated there by +2.231 dB pooled, every bin's CI excluding zero. At $P = 35$ the corresponding margin is +0.635 dB, so the model generalizes upward in pilot count far better than downward.

At 5 dB the matched-trained estimator reaches -7.52 dB at $P = 10$, within 0.47 dB of the genie-aided reference, while the $P = 20$ model evaluated there is 2.60 dB short of it. The clearest single indication that the curve is tied to its training condition is that the $P = 20$ model is worse at $P = 25$ (-11.13 dB) than at $P = 20$ (-11.27 dB): more measurements, worse estimate. Fig. 4 shows all three curves together.

This matters for how such curves are reported. The pilot sweep of [3] is described as “the NMSE performance versus the number of pilot transmissions P , evaluated at a fixed SNR of 5 dB”, and as “crucial for assessing the pilot efficiency of each algorithm”; the text does not state whether the networks were retrained per pilot count. We report that as a fact about what the paper specifies, and as motivation for stating the distinction explicitly — not as a criticism of that work, whose convention is the ordinary one and which our own earlier curves shared.

V. CONCLUSION

Per-sample normalized error over a mixed-SNR training set concentrates 89.7% of the gradient below 5 dB and leaves the high-SNR regime underfitted, so any mechanism that improves high-SNR behaviour — a structural prior among them — will appear to add information it need not have supplied. Rebalancing the loss, a change of a few lines with no architectural addition, gains +2.171 dB at $\text{SNR} \geq 5$ dB across three seeds and improves every one of eighteen bin-by-seed cells, including the two bins the weighting itself penalizes most. We recommend the matched-adequacy control be run

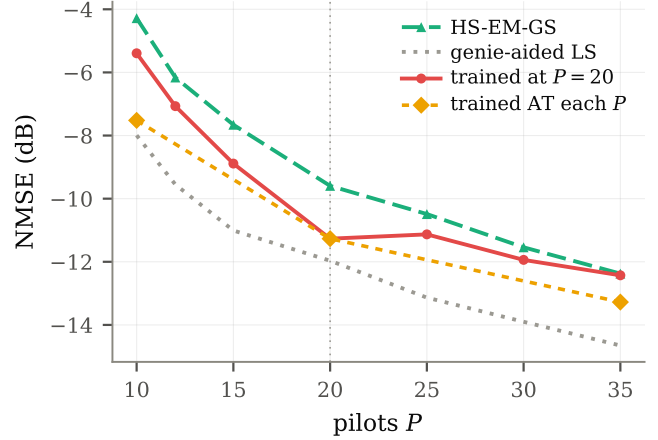


Fig. 4. Pilot efficiency against pilot-count generalization at 5 dB, on identical realizations. The dashed diamonds are separate networks trained at each P ; the solid curve is one network trained at $P = 20$.

before a structural prior is credited, and note that it cost us our own result first.

[STUB — CONCLUSION, structural-prior sentence]

One or two sentences stating what the control did to Δ_H , pending C2. The claim the retired draft made here — a fifteenfold collapse — is withdrawn and must not return in any form.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They cover the geometric ULA model of (1) at $N = 32$, $K = 3$, one data budget and the pilot counts tested; the negative result about structural priors is scoped to those conditions and is not a general claim. We establish no fundamental bound: the reference curve used throughout is genie-aided, presumes phase no receiver observes, and is passed by our own estimator at low SNR.

REFERENCES

- [1] K. Gregor and Y. LeCun, “Learning fast approximations of sparse coding,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 399–406.
- [2] V. Monga, Y. Li, and Y. C. Eldar, “Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing,” *IEEE Signal Process. Mag.*, vol. 38, no. 2, pp. 18–44, 2021.
- [3] J. Xiao, J. Wang, M. Zeng, H. Xu, X. Li, and A. Nallanathan, “Channel estimation for Rydberg atomic quantum receivers: Unrolled phase retrieval from holographic snapshots,” *IEEE Signal Process. Lett.*, vol. 33, pp. 1696–1700, 2026.
- [4] M. Cui, Q. Zeng, and K. Huang, “Towards atomic MIMO receivers,” *IEEE J. Sel. Areas Commun.*, vol. 43, no. 3, pp. 659–673, 2025.
- [5] B. Xu, J. Zhang, Z. Chen, B. Cheng, Z. Liu, Y.-C. Wu, and B. Ai, “Channel estimation for Rydberg atomic receivers,” *IEEE Wireless Commun. Lett.*, vol. 14, no. 9, pp. 2957–2961, 2025, arXiv:2503.08985.
- [6] R. W. Gerchberg and W. O. Saxton, “A practical algorithm for the determination of phase from image and diffraction plane pictures,” *Optik*, vol. 35, pp. 237–246, 1972.